Problem1

3.23. In each of the following, we specify the Fourier series coefficients of a continuous-time signal that is periodic with period 4. Determine the signal $x(t)$ in each case.

(a) $a_k = \begin{cases} 0, & k = 0 \\ (j)^k \frac{\sin k\pi/4}{k\pi}, & \text{otherwise} \end{cases}$

(b) $a_k = (-1)^k \frac{\sin k\pi/8}{2k\pi}, \quad a_0 = \frac{1}{16}$

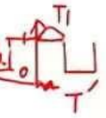
(c) $a_k = \begin{cases} jk, & |k| < 3 \\ 0, & \text{otherwise} \end{cases}$

(d) $a_k = \begin{cases} 1, & k \text{ even} \\ 2, & k \text{ odd} \end{cases}$

Problem1. $T=4 \Rightarrow \omega = \frac{2\pi}{T} = \frac{\pi}{2}$

$x(t) = \sum_{\infty} a_k e^{jk\omega t} = \sum_{\infty} a_k e^{jk\frac{\pi}{2}t}$

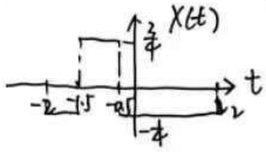
(a) $a_k = \begin{cases} 0, & k=0 \\ j^k \frac{\sin(\frac{k\pi}{4})}{k\pi}, & \text{其他} \end{cases} = e^{jk\frac{\pi}{2}} \frac{\sin(\frac{k\pi}{4})}{k\pi}$

已知方波  $a_k = \frac{\sin(k\omega_0 T_1)}{k\pi}$

$\omega T_1 = \frac{\pi}{4} \Rightarrow T_1 = \frac{1}{2}$. 对应 $g(t)$ 在 $[-0.5, 0.5]$ 上为1, 直流分量 $c_0 = \frac{2T_1}{T} = \frac{1}{4}$.

时移: $a_k \cdot e^{-jk\omega_0 t_0} \Rightarrow g(t-t_0)$

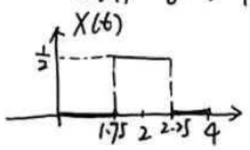
$\Rightarrow x(t) = g(t+1) - \frac{1}{4} =$



(b) $a_k = (-1)^k \frac{\sin(\frac{k\pi}{8})}{2k\pi}, \quad a_0 = \frac{1}{16}$.

$\Rightarrow a_k = \frac{1}{2} e^{jk\pi} \frac{\sin(\frac{k\pi}{8})}{k\pi} \qquad \omega_0 T_1 = \frac{\pi}{8} \Rightarrow T_1 = \frac{1}{4} \qquad -\omega t_0 = \pi \Rightarrow t_0 = -2$

直流分量 $c_0 = a_0 = \frac{1}{16}$.



(c) $a_k = \begin{cases} jk, & |k| < 3 \\ 0, & \text{其他} \end{cases}$

$x(t) = \sum_{k=-2}^{2} (jk) \cdot e^{jk\frac{\pi}{2}t} = -2j \cdot e^{-j\pi t} - je^{-j\frac{\pi}{2}t} + je^{j\frac{\pi}{2}t} + 2je^{j\pi t} = -2\sin(\frac{\pi}{2}t) - 4\sin(\pi t)$

(d) $a_k = \begin{cases} 1, & k \text{偶} \\ 2, & k \text{奇}. \end{cases} \Rightarrow a_k = 1.5 - 0.5 e^{jk\pi}$

$x(t) = \sum_{\infty} (1.5 - 0.5 e^{jk\pi}) e^{jk\frac{\pi}{2}t} = 1.5 \sum_{\infty} e^{jk\frac{\pi}{2}t} - 0.5 \sum_{\infty} e^{jk\frac{\pi}{2}(t+2)} \qquad \sum_{k=\infty} e^{jk\omega_0 t} = T \sum_{m=\infty} \delta(t - mT)$

$= 1.5 (4 \sum_{m=\infty} \delta(t-4m)) - 0.5 (4 \sum_{m=\infty} \delta(t+2-4m))$

$= \sum_{m=-\infty}^{\infty} [6\delta(t-4m) - 2\delta(t-4m-2)]$

Problem2

3.41. Suppose we are given the following information about a continuous-time periodic signal with period 3 and Fourier coefficients a_k:

1. $a_k = a_{k+2}$.
2. $a_k = a_{-k}$.
3. $\int_{-0.5}^{0.5} x(t)dt = 1$.
4. $\int_1^2 x(t)dt = 2$.

Determine $x(t)$.

Problem2 $T=3 \Rightarrow \omega = \frac{2\pi}{T} = \frac{2\pi}{3}$.

令 $a_k = \begin{cases} A, & k\text{偶} \\ B, & k\text{奇} \end{cases}$ $\quad x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega t} = \sum_{m=-\infty}^{\infty} \left[A e^{j\cdot 2m\cdot\frac{2\pi}{3}t} + B e^{j(2m+1)\frac{2\pi}{3}t} \right]$

$$= \sum_{m=-\infty}^{\infty} e^{jm\frac{4\pi}{3}t} \left(A + B e^{j\frac{2\pi}{3}t}\right)$$

又 $\sum_{m=-\infty}^{\infty} e^{jm\frac{4\pi}{3}t} = \frac{2\pi}{\frac{4\pi}{3}} \sum_{n=-\infty}^{\infty} \delta\left(t - n\frac{2\pi}{\frac{4\pi}{3}}\right)$

$$= 1.5 \sum_{n=-\infty}^{\infty} \delta(t-1.5n)$$

$\Rightarrow x(t) = 1.5\left(A + B e^{j\frac{2\pi}{3}t}\right) \sum_{n=-\infty}^{\infty} \delta(t-1.5n) = 1.5 \sum_{n=-\infty}^{\infty} \left[A + B(-1)^n\right] \delta(t-1.5n)$ 每隔1.5s出现一次冲激

$$\begin{cases} \int_{-0.5}^{0.5} x(t)dt = 1 \Rightarrow n=0,\ 1.5(A+B)=1 \\ \int_1^2 x(t)dt = 2 \Rightarrow t=1.5,\ n=1,\ 1.5(A-B)=2 \end{cases} \Rightarrow \begin{cases} A=1 \\ B=-\frac{1}{3} \end{cases}$$

$1.5\left(1-\frac{1}{3}\right)=1, \quad 1.5\left(1+\frac{1}{3}\right)=2$

$\Rightarrow x(t) = \sum_{k=-\infty}^{\infty} \left[\delta(t-3k) + 2\delta(t-3k-1.5)\right]$

Problem3

3.44. Suppose we are given the following information about a signal $x(t)$:

1. $x(t)$ is a real signal.
2. $x(t)$ is periodic with period $T = 6$ and has Fourier coefficients a_k.
3. $a_k = 0$ for $k = 0$ and $k > 2$.
4. $x(t) = -x(t-3)$.
5. $\frac{1}{6}\int_{-3}^{3}|x(t)|^2\,dt = \frac{1}{2}$.
6. a_1 is a positive real number.

Show that $x(t) = A\cos(Bt + C)$, and determine the values of the constants A, B, and C.

Problem3 $T=6 \Rightarrow \omega = \frac{2\pi}{T} = \frac{\pi}{3}$

$x(t)$是实信号 $\Rightarrow a_{-k} = a_k^*$ } 只有 a_1, a_{-1}, a_2, a_{-2} 可能不为0.

$k \geq 0$时只有 a_1 a_2 不为0

$x(t) = -x(t-3) \Rightarrow a_k = a_k \cdot e^{-jk\frac{\pi}{3}\cdot 3} \Rightarrow a_k = -a_k(-1)^k$

$\begin{cases} k偶: a_k = -a_k,\ a_k = 0 \\ k奇: 恒成立 \end{cases} \Rightarrow$ 只有 a_1, a_{-1} 可能不为0.

$\frac{1}{6}\int_{-3}^{3}|x(t)|^2 dt = \frac{1}{2} \Rightarrow \frac{1}{T}\int_T |x(t)|^2 dt = \sum_{k=-\infty}^{\infty}|a_k|^2 = \frac{1}{2} = |a_1|^2 + |a_{-1}|^2$

又 $a_1 = a_1^* \Rightarrow |a_1| = |a_{-1}| = \frac{1}{2}$

又 a_1 是正实数 $\Rightarrow a_1 = a_{-1} = \frac{1}{2} \Rightarrow x(t) = \frac{1}{2}(e^{j\frac{\pi}{3}t} + e^{-j\frac{\pi}{3}t}) = \cos\frac{\pi}{3}t \Rightarrow \begin{cases} A=1 \\ B=\frac{\pi}{3} \\ C=0 \end{cases}$

Problem4

4.12. Consider the Fourier transform pair

$$e^{-|t|} \stackrel{\mathcal{F}}{\longleftrightarrow} \frac{2}{1+\omega^2}.$$

(a) Use the appropriate Fourier transform properties to find the Fourier transform of $te^{-|t|}$.

(b) Use the result from part (a), along with the duality property, to determine the Fourier transform of

$$\frac{4t}{(1+t^2)^2}.$$

Hint: See Example 4.13.

Problem4 已知 $e^{-|t|} \xrightarrow{F} \frac{1}{1+\omega^2}$

(1) 则 $te^{-|t|} \xrightarrow{F} j\frac{d}{d\omega}\left(\frac{1}{1+\omega^2}\right) = \frac{-j4\omega}{(1+\omega^2)^2}$

(2) 对偶性质: $g(t) \xrightarrow{F} f(\omega)$ 则 $f(t) \xrightarrow{F} 2\pi g(-\omega)$

则 $\frac{4t}{(1+t^2)^2} \xrightarrow{F} \frac{2\pi}{-j}(-\omega)e^{-|\omega|} = -j2\pi\omega e^{-|\omega|}$

Problem5

4.24. **(a)** Determine which, if any, of the real signals depicted in Figure P4.24 have Fourier transforms that satisfy each of the following conditions:

(1) $\Re e\{X(j\omega)\} = 0$

(2) $\Im m\{X(j\omega)\} = 0$

(3) There exists a real α such that $e^{j\alpha\omega}X(j\omega)$ is real

(4) $\int_{-\infty}^{\infty} X(j\omega)\,d\omega = 0$

(5) $\int_{-\infty}^{\infty} \omega X(j\omega)\,d\omega = 0$

(6) $X(j\omega)$ is periodic

(b) Construct a signal that has properties (1), (4), and (5) and does *not* have the others.

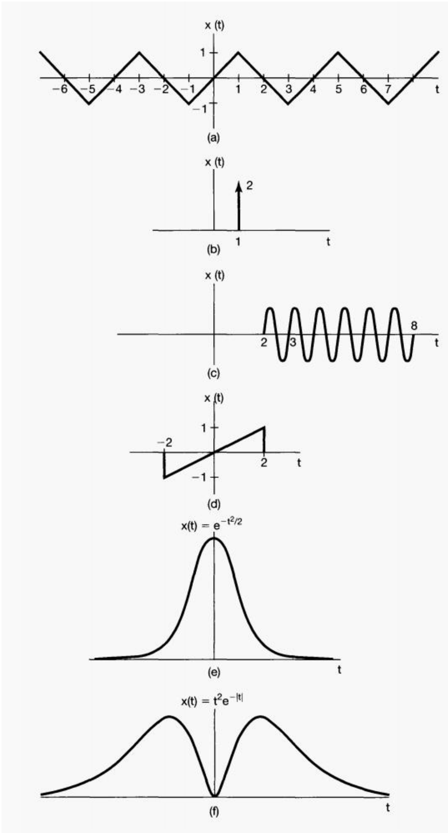


Figure P4.24

$X(j\omega) = \int_{\infty} x(t)e^{-j\omega t}dt$

$= \int_{\infty} x(t)[\cos\omega t - j\sin\omega t]dt$

实部$=0 \Rightarrow \int_{\infty} x(t)\cos\omega t = 0 \Rightarrow x(t)$奇.

Problem5 (1) ① $Re\{X(j\omega)\}=0 \Rightarrow x(t)$奇函数

② $Im\{X(j\omega)\}=0 \Rightarrow x(t)$偶函数

③ 存在a使$e^{ja\omega}X(j\omega)$为实数：$F(x(t+a))$实数 $\Rightarrow x(t+a)$偶函数，$x(t)$存在对称轴.

④ $\int_{\infty} X(j\omega)d\omega = 0 \Rightarrow x(0) = \frac{1}{2\pi}\int_{\infty} X(j\omega)d\omega = 0$.

⑤ $\int_{\infty} \omega X(j\omega)d\omega = 0 \Rightarrow x'(0) = \frac{j}{2\pi}\int_{\infty}\omega X(j\omega)d\omega = 0$.

⑥ $X(j\omega)$周期 $\Rightarrow x(t)$由冲激信号组成.

①: ad ②: ef ③ abef ④ abcdf ⑤ bcef ⑥ b.

(2) $x(t) = t^3 \cdot e^{-t^2}$ ($x(t)=t^3 \to \infty$ 不可积)

Problem6

4.33. The input and the output of a stable and causal LTI system are related by the differential equation

$$\frac{d^2y(t)}{dt^2} + 6\frac{dy(t)}{dt} + 8y(t) = 2x(t)$$

(a) Find the impulse response of this system.
(b) What is the response of this system if $x(t) = te^{-2t}u(t)$?
(c) Repeat part (a) for the stable and causal LTI system described by the equation

$$\frac{d^2y(t)}{dt^2} + \sqrt{2}\frac{dy(t)}{dt} + y(t) = 2\frac{d^2x(t)}{dt^2} - 2x(t)$$

Problem 6. (1) $y''(t) + 6y'(t) + 8y(t) = 2x(t)$

$\Rightarrow (j\omega)^2 Y(j\omega) + 6j\omega \cdot Y(j\omega) + 8Y(j\omega) = 2X(j\omega)$

$\Rightarrow H(j\omega) = \frac{2}{(j\omega)^2 + 6j\omega + 8} = \frac{1}{j\omega+2} - \frac{1}{j\omega+4} \quad \Rightarrow h(t) = (e^{-2t} - e^{-4t})u(t)$

(2) $x(t) = te^{-2t}u(t)$, $X(j\omega) = \frac{1}{(j\omega+2)^2}$

$\Rightarrow Y(j\omega) = \frac{2}{(j\omega+2)^3(j\omega+4)} = \frac{1}{(j\omega+2)^3} - \frac{0.5}{(j\omega+2)^2} + \frac{0.25}{j\omega+2} - \frac{0.25}{j\omega+4}$

$\Rightarrow y(t) = [\frac{1}{2}t^2e^{-2t} - \frac{1}{2}te^{-2t} + \frac{1}{4}e^{-2t} - \frac{1}{4}e^{-4t}]u(t)$

(3) $y''(t) + \sqrt{2}y'(t) + y(t) = 2x''(t) - 2x(t)$

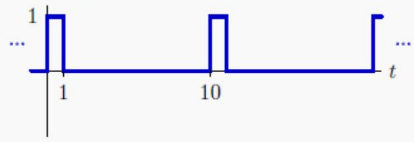
$\Rightarrow (s^2+\sqrt{2}s+1)Y(s) = (2s^2-2)X(s)$, $H(s) = \frac{2s^2-2}{s^2+\sqrt{2}s+1} = 2 - 2\sqrt{2}\left[\frac{s+\frac{\sqrt{2}}{2}}{(s+\frac{\sqrt{2}}{2})^2+\frac{1}{2}} + \frac{\frac{\sqrt{2}}{2}}{(s+\frac{\sqrt{2}}{2})^2+\frac{1}{2}}\right]$

$\Rightarrow h(t) = 2\delta(t) - 2\sqrt{2}e^{-\frac{\sqrt{2}}{2}t}[\cos(\frac{\sqrt{2}}{2}t) + \sin(\frac{\sqrt{2}}{2}t)]u(t)$.

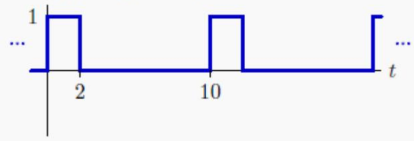
Problem 7：

Determine the Fourier series coefficients for each of the following periodic CT signals.

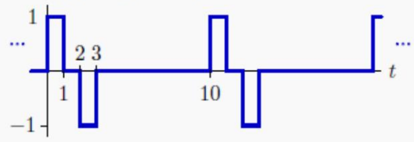
$x_1(t)= x_1(t+10)$



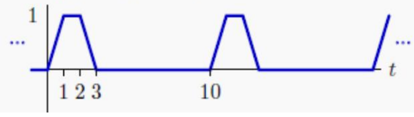
$x_2(t)= x_2(t+10)$



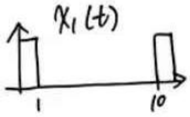
$x_3(t)= x_3(t+10)$



$x_4(t)= x_4(t+10)$



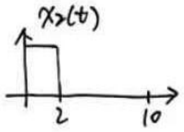
Problem7. (1)



$$T=10 \Rightarrow \omega=\frac{\pi}{5}$$

$$a_k=\frac{\sin k\cdot\frac{\pi}{5}\cdot\frac{1}{2}}{k\pi}e^{-jk\frac{\pi}{5}\cdot\frac{1}{2}}=\frac{\sin\frac{k\pi}{10}}{k\pi}e^{-jk\frac{\pi}{10}}, \quad a_0=\frac{1}{10}$$

(2)



$$\omega=\frac{\pi}{5}$$

$$a_k=\frac{\sin\frac{k\pi}{5}}{k\pi}e^{-jk\frac{\pi}{5}}, \quad a_0=\frac{1}{5}$$

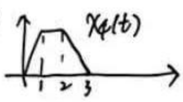
(3) $x_3(t)=x_1(t)-x_1(t-2)$

$$a_k=a_{1k}\left(1-e^{-jk\frac{\pi}{5}\cdot 2}\right)$$

$$=\frac{\sin\frac{k\pi}{10}}{k\pi}\cdot e^{-jk\frac{\pi}{5}}\left(2j\cdot\sin\frac{k\pi}{5}\right)=\frac{2j}{k\pi}\sin\frac{k\pi}{10}\sin\frac{k\pi}{5}e^{-jk\frac{3\pi}{10}}$$

$$a_0=0$$

(4)



$$\frac{dx_4(t)}{dt}=x_3(t)$$

$$\Rightarrow a_k=\frac{a_{3k}}{jk\frac{\pi}{5}}=\frac{10}{k^2\pi^2}\sin\frac{k\pi}{10}\sin\frac{k\pi}{5}e^{-jk\frac{3\pi}{10}}, \quad a_0=\frac{1}{5}$$

Problem 8:

Find the Fourier transforms of the following signals.

a. $x_1(t) = e^{-|t|}\cos(2t)$

b. $x_2(t) = \dfrac{\sin(2\pi t)}{\pi(t-1)}$

c. $x_3(t) = \begin{cases} t^2 & 0 < t < 1 \\ 0 & \text{otherwise} \end{cases}$

d. $x_4(t) = (1-|t|)\,u(t+1)u(1-t)$

Problem8 (a) $x_1(t) = e^{-|t|}\cos 2t.$

$e^{-a|t|} \xrightarrow{F} \dfrac{2a}{a^2+\omega^2}$

$x(t)\cos(\omega_0 t) \xrightarrow{F} \frac{1}{2}X(\omega-\omega_0) + \frac{1}{2}X(\omega+\omega_0)$

$e^{-|t|} \xrightarrow{F} \dfrac{2}{1+\omega^2}$

$x_1(t) \xrightarrow{F} \dfrac{1}{2}\left[\dfrac{2}{1+(\omega-2)^2} + \dfrac{2}{1+(\omega+2)^2}\right]$

$\Rightarrow X_1(\omega) = \dfrac{1}{1+(\omega-2)^2} + \dfrac{1}{1+(\omega+2)^2}$

(b) $x_2(t) = \dfrac{\sin(2\pi t)}{\pi(t-1)}$

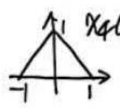
对 $g(t) = \dfrac{\sin 2\pi t}{\pi t}$, $G(\omega) = \begin{cases} 1, & |\omega| < 2\pi \\ 0, & |\omega| > 2\pi \end{cases}$

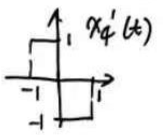
$x_2(t) = \dfrac{\sin 2\pi(t-1)}{\pi(t-1)} = g(t-1)$, $X_2(\omega) = G(\omega)\cdot e^{-j\omega} = \begin{cases} e^{-j\omega}, & |\omega| < 2\pi \\ 0, & |\omega| > 2\pi. \end{cases}$

(c) $x_3(t) = \begin{cases} t^2, & 0<t<1 \\ 0, & \text{其他}. \end{cases}$

$X_3(\omega) = \int_0^1 t^2 e^{-j\omega t}\,dt = \left[e^{-j\omega t}\left(\dfrac{t^2}{-j\omega} - \dfrac{2t}{(-j\omega)^2} + \dfrac{2}{(-j\omega)^3}\right)\right]_0^1 = e^{-j\omega}\left(\dfrac{j}{\omega} + \dfrac{2}{\omega^2} - \dfrac{2j}{\omega^3}\right) + \dfrac{2j}{\omega^3}$

(d) $x_4(t) = (1-|t|)\,u(t+1)u(1-t)$





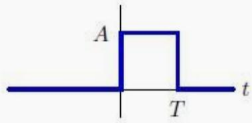
$x_4''(t) = \delta(t+1) - 2\delta(t) + \delta(t-1)$

$F(x_4''(t)) = e^{j\omega} - 2 + e^{-j\omega} = 2\cos\omega - 2 = -4\sin^2\left(\frac{\omega}{2}\right)$

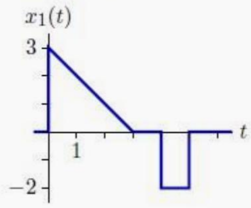
$\cancel{X_4(\omega) = (j\omega)^2 X}$ $F(x_4''(t)) = (j\omega)^2 X_4(\omega) \Rightarrow X_4(\omega) = \dfrac{4\left(\sin\frac{\omega}{2}\right)^2}{\omega^2}$

Problem 9：

We are given that the impulse response of a CT LTI system is of the form



where A and T are unknown. When the system is subjected to the input

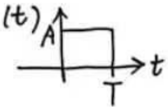


the output $y_1(t)$ is zero at $t = 5$. When the input is

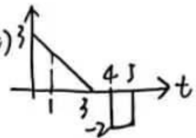
$$x_2(t) = \sin\left(\frac{\pi t}{3}\right) u(t),$$

the output $y_2(t)$ is equal to 9 at $t = 9$. Determine A and T. Also determine $y_2(t)$ for all t.

Problem9 已知 $h(t)$



$\begin{cases} x_1(t) \\ x_2(t) = \sin(\frac{\pi t}{3})u(t) \Rightarrow y_2(t): y(9)=9 \end{cases}$



$\rightarrow y_1(t)$: ~~$y_1(t)=0$ $y_1(t)$在$t=5$时为0~~

$y_1(5)=0$

$y_1(5) = A\int_{5-T}^{5} x_1(\tau)\,d\tau = 0 \Rightarrow T=4$

$y_2(9) = A\int_{9-4}^{9} \sin(\frac{\pi\tau}{3})\,d\tau = A\cdot[-\frac{3}{\pi}\cos\frac{\pi\tau}{3}]_5^9 = A\cdot\frac{9}{2\pi} = 9 \Rightarrow A=2\pi$

$\Rightarrow \begin{cases} T=4 \\ A=2\pi. \end{cases}$

$y_2(t) = 2\pi\int_0^t \sin(\frac{\pi\tau}{3})[u(t-\tau)-u(t-\tau-4)]\,d\tau$

① $t<0$时, $y_2(t)=0$

② $0<t<4$时, $y_2(t) = 2\pi\int_0^t \sin(\frac{\pi\tau}{3})\,d\tau = 6-6\cos\frac{\pi t}{3}$

③ $t>4$时, $y_2(t) = 2\pi\int_{t-4}^t \sin(\frac{\pi\tau}{3})\,d\tau = -6\sqrt{3}\sin(\frac{\pi t}{3}-\frac{2\pi}{3})$